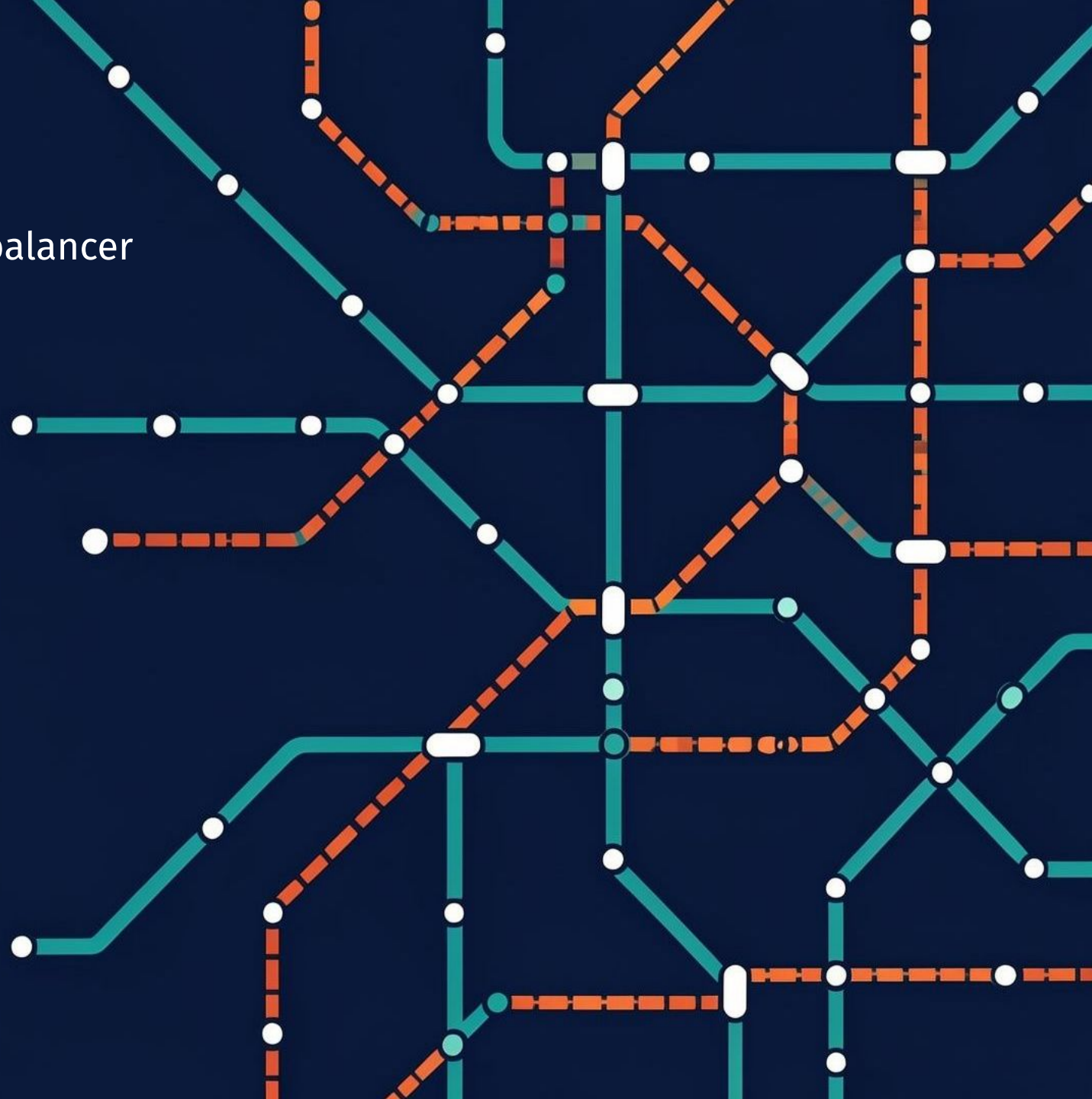
Frustrum

AI/NLP Risk Engine and Tactical Index Rebalancer

GARLAPATI GAGANA · IIIT LUCKNOW





Headlines should update belief, not dictate weight

Markets move on news.

News is noisy, people overreact, and every trade has a cost.

Views stay noisy.

Frustrum turns NLP output into small, bounded, costed trades.

Weights earn their size.

“We do not let a headline set a weight. We let it set a belief, then size the trade by confidence, decay and cost.”

Reader → NLP Risk Engine · Decision maker → Tactical Rebalancer · Scoreboard → Dashboard and backtest



SYSTEM MAP

From news text to explainable, cost-aware trades

A left-to-right path from evidence to bounded action.



News text

Demo news is synthetic and labelled.



NLP Risk Engine

FinBERT sentiment · MiniLM event classification · rule-based impact 1–10.



SQLite + FastAPI

Stores views and serves the decision layer.



Rebalancer

Per-stock views, exponential tilt, SLSQP optimisation, and guardrails.



Dashboard + replay

Streamlit dashboard and replay backtest; wired to yfinance daily prices, with the measured run pending.

15-stock mock index · equal-weight benchmark at 6.67% each · AAPL, MSFT, NVDA, GOOGL, AMZN, JPM, BAC, XOM, CVX, JNJ, PFE, WMT, KO, BA, DIS. Engine: ticker linking, novelty/deduplication, and confidence. yfinance daily prices feed correlation and covariance.

DESIGN CHOICES

Six design choices make the signal usable



Signal fusion

Sentiment × source weight × impact × confidence × novelty × time decay.



Time and novelty

Earnings fade in 1 day; credit events in 10 days. Near-duplicates get novelty 0; contradictory social is halved.



Contagion

Views spill to correlated peers; macro and geopolitical stories spread to sectors.



Optimise

Tilt the benchmark under weight bounds and a 35% sector cap.



Explain

“Why Changed” links moves to headlines.



Prove

Ablations and cost sensitivity test the design. Impact is transparent and rule-based: event base score + sentiment strength + severity keywords.

EVIDENCE STATUS

Results are ready for evidence, not invention

The pipeline is prepared to show performance; final backtest metrics and equity-curve output await the measured run.

Equity curves

Frustrum · equal-weight hold · equal-weight daily rebalance · naive sentiment tilt



Signals published before the bar · metrics on the last 40% of the timeline · 5 bps on every rebalance

Metrics table

METRIC	VALUE
Annualised return	PENDING MEASURED RUN
Volatility	PENDING MEASURED RUN
Sharpe	PENDING MEASURED RUN
Max drawdown	PENDING MEASURED RUN
Annual turnover	PENDING MEASURED RUN
Mean IC	PENDING MEASURED RUN

ABLATION BARS

No decay · no dedup · no contagion · no cost control

PENDING MEASURED RUN

COST SENSITIVITY

0 · 5 · 10 · 25 bps

PENDING MEASURED RUN

Baselines included. Synthetic news validates the pipeline, not real-world alpha; GDELT and tweet/social data are planned, not yet verified.

RISK CONTROLS

A desk gets a weak signal with strong guardrails

The system can surface risk without turning one headline into a concentrated bet.

Position limits

Weight band 2%–12%; max tilt ±50% of benchmark; sector cap 35%.

Turnover cap 10% per rebalance · 5 bps transaction cost · 0.5% no-trade band

Emergency logic

Cooldown: 15 minutes.

Circuit breaker: impact ≥ 9, sentiment ≤ −0.6, confidence ≥ 0.5.

Pin the stock at the 2% floor for 60 minutes.

Desk benefit

Less manual news triage, event-driven risk surfaced, explainable allocation changes, transparent limits.

DASHBOARD SCREENSHOT · PENDING MEASURED RUN

Weights chart · “Why Changed” headline link · circuit-breaker marker

Awaiting the measured run; insert the Streamlit capture before judging the live demo.

CONTROL PATH

Signal → constraint → allocation → explanation

FRUSTRUM

Limitations & Next Steps

The boundary between demo and evidence

Assumptions I chose

Half-lives by event type, a sector sensitivity table, an assumed 10 bps return per unit of view, a 5 bps fee, and source trust levels.

Tuned on the training window only.

Data limits

Short sample of large US stocks; English-only FinBERT can misread sarcasm; free feeds have gaps and delays; no real orders or slippage.

Demo news is synthetic. GDELT and tweet/social data are intended real-source next steps—not completed evidence.

Next steps

Fine-tune on labelled finance text; add volume and intraday signals; paper-trade live; add more sources; swap SQLite for a queue to scale.



Weak signal, strong guardrails: here is what we proved and what we didn’t.